Adaptive Fraud Detection with Dynamic Thresholding and Cost-Sensitive Learning

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***Abstract*—We present a complete supervised machine learning pipeline for credit card fraud detection on the Kaggle Credit Card Fraud Detection dataset (284,807 transactions; 0.17% fraud prevalence). Three classifiers are trained and compared: Logistic Regression (LR), Random Forest (RF), and Histogram-based Gradient Boosted Trees (GBT). Three class-imbalance strategies are evaluated: balanced class weighting, SMOTE, and random under-sampling. We introduce dynamic threshold tuning on the validation set and a cost-sensitive evaluation framework that minimises expected business cost, modelling missed fraud at €500 and a false alarm at €10 per transaction. The best model — RF with balanced class weights at a tuned threshold of 0.192 — achieves PR-AUC = 0.8116, Recall = 82.4%, and Precision = 84.7% on the held-out test set. Code and notebook are available at: https://github.com/omsureshprajapati/EE559-Fraud-Detection.**

***Index Terms*—fraud detection, class imbalance, SMOTE, threshold tuning, cost-sensitive learning, random forest, gradient boosted trees**

I. Introduction

Credit card fraud causes billions of euros in annual losses to financial institutions [1]. Automated detection reduces this burden, but the problem presents two fundamental challenges. First, the extreme class imbalance (577:1 in this dataset) makes standard accuracy uninformative: a classifier that labels every transaction as legitimate achieves 99.83% accuracy while catching zero fraud. Second, the two error types carry asymmetric costs: a missed fraud (false negative, FN) costs far more than a false alarm (false positive, FP) that triggers a manual review. Any practical system must account for this asymmetry explicitly.

This paper addresses both challenges. We use Precision-Recall AUC (PR-AUC) and ROC-AUC as primary metrics [2], compare three class-imbalance handling strategies, and introduce a cost-sensitive threshold selection procedure that finds the operating point minimising total expected business cost on the validation set.

II. Dataset and Preprocessing

A. Dataset

The Kaggle Credit Card Fraud Detection dataset [3] contains 284,807 European cardholder transactions from September 2013, with 492 fraud cases (0.173%). Features `V1–V28` are anonymised PCA components (zero-mean, decorrelated); `Time` (seconds since first transaction) and `Amount` (EUR) are in original units. No missing values were present; 1,081 duplicate rows were retained.

B. Preprocessing Pipeline

The following strict ordering prevents data leakage:

1) **Split first.** Stratified 70/15/15 train/validation/test split (199,356 / 42,729 / 42,722 samples), preserving 0.173% fraud prevalence in each subset. The test set is held out until Section IV-D.
2) **Scale on train only.** `StandardScaler` fitted on `Time` and `Amount` from the training data; applied to all splits. Features `V1–V28` require no re-scaling (already zero-mean from PCA).
3) **Resample train only.** SMOTE and under-sampling applied within the training split only; validation and test sets are never modified.

III. Methodology

A. Models

Logistic Regression (LR). Linear baseline optimising cross-entropy with L2 regularisation ($C = 1.0$):

$$P(y{=}1 \mid \mathbf{x}) = \sigma\big(\mathbf{w}^\top \mathbf{x} + b\big)\,, \quad \mathcal{L} = \mathrm{CE} + \tfrac{1}{2C}\|\mathbf{w}\|_2^2. \qquad (1)$$

`class_weight='balanced'` reweights each fraud sample by $\approx 577\times$ relative to legitimate samples.

Random Forest (RF). Ensemble of 100 decision trees, each trained on a bootstrap sample with random feature subsets at each split [4]. Captures nonlinear feature interactions. Same balanced class weighting.

Gradient Boosted Trees (GBT). `HistGradientBoostingClassifier` [6] (200 iterations). Additively builds shallow trees correcting ensemble residuals. Trained in two variants: balanced class weighting (GBT-bal) and SMOTE-resampled training data (GBT-SMOTE).

TABLE I: Resampling strategy comparison on LR (validation, threshold = 0.50).

Strategy	PR-AUC	Recall	Fraud F1
Class weight	0.7520	0.9189	0.1265
SMOTE	0.7599	0.9324	0.1209
Under-sampling	0.6986	0.9324	0.0865

B. Class Imbalance Handling

Three strategies were compared using LR to isolate resampling effects: (1) `class_weight='balanced'`; (2) SMOTE [5], which interpolates between minority samples to create synthetic fraud examples, bringing the training ratio to 1:1; and (3) Random under-sampling, which discards majority samples until 1:1.

C. Threshold Tuning

For threshold t, the predicted class is $\hat{y} = \mathbf{1}[\hat{p} \geq t]$. We sweep $t \in [0.01, 0.99]$ (200 steps) on the validation set and identify:

- **F1-optimal:** $t^*_{\text{F1}} = \arg\max_t \text{F1}(t)$
- **Cost-optimal:** $t^*_{\text{cost}} = \arg\min_t \mathcal{C}(t)$

where the business cost function is:

$$\mathcal{C}(t) = \text{FN}(t) \cdot C_{\text{FN}} + \text{FP}(t) \cdot C_{\text{FP}} \tag{2}$$

with $C_{\text{FN}} = 500$ € (average fraud transaction loss) and $C_{\text{FP}} = 10$ € (manual review cost). All threshold selection uses only the validation set.

IV. Results

A. Resampling Strategy Comparison

Table I shows LR performance under the three imbalance strategies at threshold 0.50. Class weighting and SMOTE achieve similar PR-AUC; random under-sampling underperforms both because it discards 199,012 majority-class training samples (99.8% of legitimate transactions), losing discriminative signal. SMOTE is therefore applied to GBT as the secondary variant.

B. Model Comparison at F1-Optimal Threshold

Table II compares all model variants at their F1-optimal thresholds on the validation set. RF achieves the highest PR-AUC (0.8637), confirming that nonlinear feature interactions are important for fraud discrimination. GBT-SMOTE achieves the highest F1 (0.8794) but at a lower PR-AUC (0.8533), indicating it is more sensitive to threshold choice. LR lags substantially on both metrics.

The F1-optimal thresholds diverge widely across models: RF's optimum is 0.192 (far below the default 0.50) because its probabilities are well-calibrated and the fraud class is extremely rare. LR requires a threshold of 0.990 because its raw fraud probabilities are small in absolute terms.

TABLE II: All models at F1-optimal thresholds (validation set). Best in **bold**.

Model	PR-AUC	ROC-AUC	Thr.	F1	Rec.	Prec.
LR (bal.)	0.752	0.986	0.990	0.727	0.865	0.628
RF (bal.)	**0.864**	0.951	**0.192**	0.877	**0.865**	0.889
GBT (bal.)	0.784	**0.982**	0.951	0.853	0.824	0.884
GBT (SMOTE)	0.853	0.973	0.960	**0.879**	0.838	**0.925**

TABLE III: F1-optimal vs. cost-optimal thresholds (validation set).

Model	F1-opt. Thr.	Cost-opt. Thr.
LR (balanced)	0.990	0.985
RF (balanced)	0.192	0.044
GBT (balanced)	0.951	0.837
GBT (SMOTE)	0.960	0.251

C. Threshold Tuning and Cost Analysis

Fig. 1 shows F1, Precision, and Recall as functions of the decision threshold for all models. Fig. 2 shows the total business cost from (2) vs. threshold.

Because $C_{\text{FN}} \gg C_{\text{FP}}$ (50:1 ratio), the cost-optimal thresholds are substantially lower than the F1-optimal thresholds for all models: the system prefers flagging more transactions (accepting more FP) to avoid missing fraud (reducing FN). For RF, the cost-optimal threshold drops from 0.192 to 0.044, catching one additional fraud (FN: 13 → 12) while increasing FP from 11 to 48, yielding a net saving of €130. Table III summarises the thresholds.

D. Final Test Set Evaluation

RF (balanced) was selected as the best model by validation PR-AUC = 0.8637. The held-out test set (74 fraud; 42,648 legitimate) was evaluated exactly once. Table IV reports results at both thresholds. Fig. 3 shows the confusion matrix, ROC curve, and PR curve.

At the F1-optimal threshold the model catches 61 of 74 frauds (82.4% recall) with only 11 false alarms. The 13 missed frauds represent €6,500 in FN losses; the 11 false alarms add €110 in review costs (total: €6,610).

V. Error Analysis

The 13 false negatives (missed frauds at threshold 0.192) had a mean model score of 0.021, compared to 0.826 for correctly detected frauds. Feature distribution analysis of the top discriminative features (`V14`, `V17`, `V12`) shows that missed frauds have values close to the legitimate class distribution, indicating they are genuinely ambiguous transactions rather than model failures. Their mean transaction amount was lower than detected frauds, consistent with small test charges being harder to distinguish from low-value legitimate transactions.

The 11 false positives had a mean score of 0.44, just above the decision threshold, suggesting legitimate transactions with unusual PCA projections that resemble known fraud signatures.

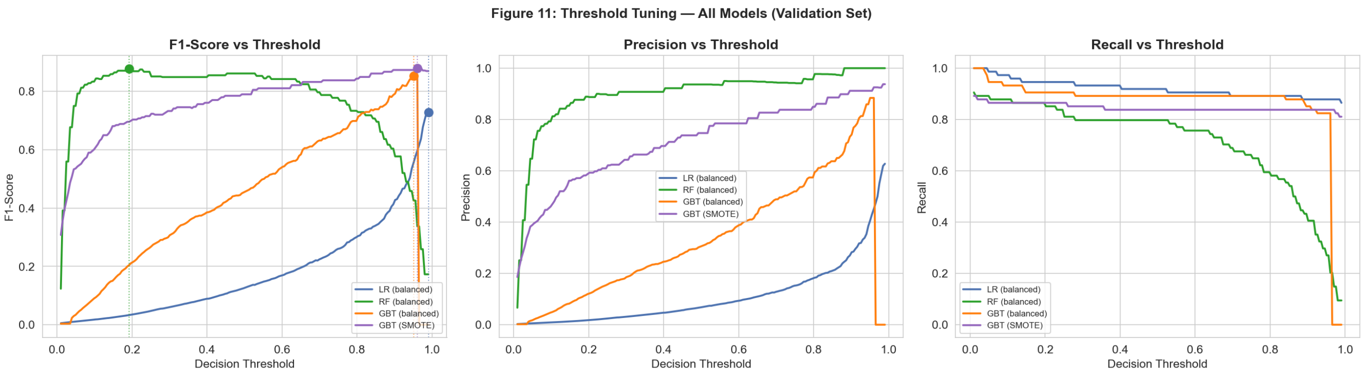

Fig. 1: F1-score, Precision, and Recall vs. decision threshold for all four models on the validation set. Dots mark the F1-maximising threshold per model. Note the wide variation in optimal thresholds across model families.

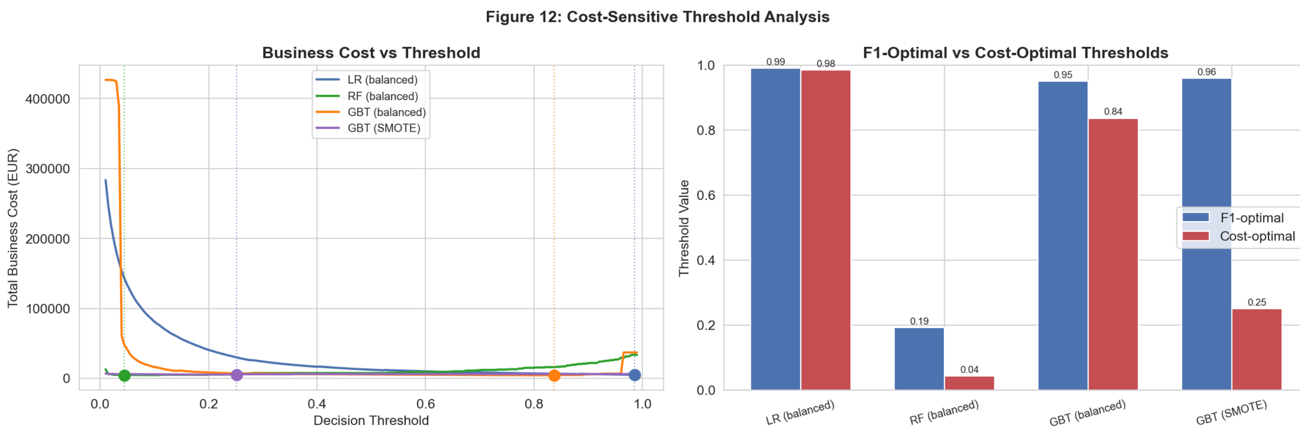

Fig. 2: Left: total business cost (€500 × FN + €10 × FP) vs. threshold for all models (validation set); dots mark cost-minimising thresholds. Right: F1-optimal vs. cost-optimal thresholds per model — cost-optimal thresholds are consistently lower due to the 50:1 FN/FP penalty.

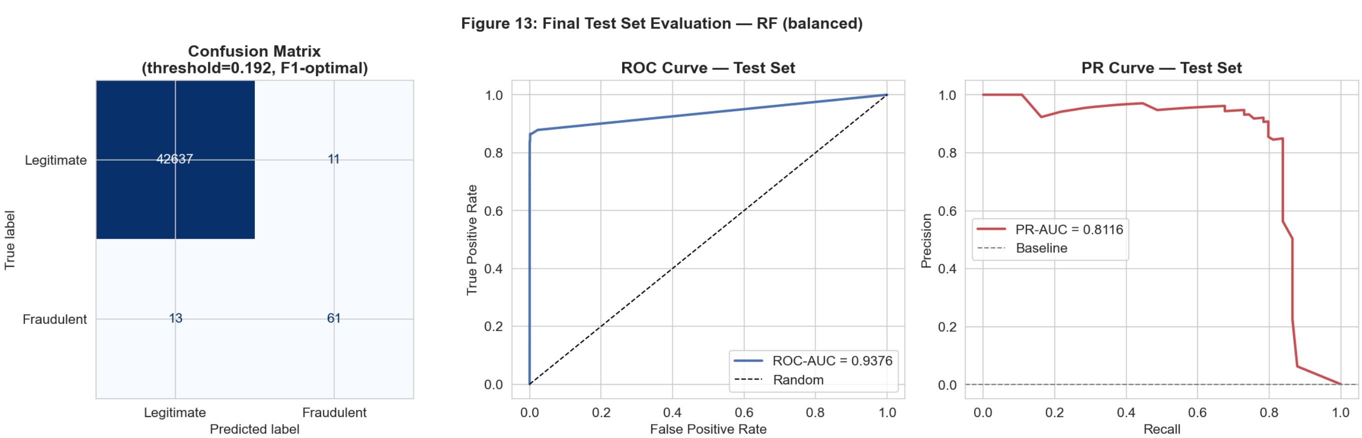

Fig. 3: Final test-set evaluation for RF (balanced) at threshold 0.192. Left: confusion matrix (61 TP, 11 FP, 13 FN, 42,637 TN). Centre: ROC curve (AUC = 0.9376). Right: Precision-Recall curve (AUC = 0.8116).

VI. Discussion

Nonlinear models are essential. RF and GBT outperform LR by 0.11 PR-AUC, confirming that fraud signals arise from nonlinear feature combinations that a linear boundary cannot capture.

Threshold tuning is as important as model choice. Tuning the RF threshold from 0.50 to 0.192 improves validation F1

TABLE IV: RF (balanced) on held-out test set (74 fraud; 42,648 legitimate).

Threshold	ROC	PR	F1	Rec.	Prec.	Cost (€)
0.192 (F1-opt.)	0.9376	0.8116	0.8356	0.8243	0.8472	6,610
0.044 (cost-opt.)	0.9376	0.8116	0.6739	0.8378	0.5636	6,480

from 0.8613 to 0.8767 with no model change — a gain equal to the difference between RF and LR.

Cost-sensitive analysis changes the operating point. The 50:1 FN/FP cost ratio shifts the RF optimal threshold from 0.192 to 0.044. Whether this trade-off is operationally acceptable depends on the institution's manual review capacity.

SMOTE vs. class weighting. On LR, SMOTE achieves marginally higher PR-AUC (+0.008). On GBT, class weighting yields higher PR-AUC while SMOTE yields higher F1, suggesting class weighting is better calibrated across all thresholds.

A. Limitations

Features `V1`–`V28` are opaque PCA components with no semantic interpretation. Hyperparameter tuning (grid search over tree depth, learning rate, number of estimators) was not performed. The 48-hour observation window limits temporal generalisation to evolving fraud strategies.

B. Future Work

Time-aware cross-validation (respecting transaction ordering), hyperparameter optimisation, SHAP-based explanations [7], and online learning [8] to handle distribution shift remain as extensions.

VII. Conclusion

We presented a complete fraud detection pipeline comparing three classifiers, three imbalance-handling strategies, dynamic threshold tuning, and cost-sensitive evaluation. Random Forest with balanced class weights and a tuned threshold of 0.192 is the best model, achieving PR-AUC = 0.8116 and F1 = 0.8356 on the test set. A key finding is that threshold selection under asymmetric business costs yields a meaningfully different operating point than F1 maximisation, highlighting the importance of tying model evaluation to concrete cost assumptions.

References

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